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TREND-BASED MOCK PAPER

GATE Civil Engineering

Mock Paper — 02

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Civil Engineering
Paper:	Mock 02
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

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Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section A: General Aptitude

Q1. [1-MCQ] [1 mark]

Volume of sphere r:

- (A) $\frac{4}{3} \pi r^3$
- (B) $\frac{4}{3} \pi r^2$
- (C) $\frac{4}{3} \pi r$
- (D) $2\pi r$

Q2. [1-MCQ] [1 mark]

Choose similar to 'diligent':

- (A) Lazy
- (B) Industrious
- (C) Careless
- (D) Ignorant

Section B: Civil Engineering

Q3. [1-MCQ] [1 mark]

Order of $\frac{dy}{dx}=y$:

- (A) 0
- (B) 1
- (C) 2
- (D) 3

Q4. [2-NAT] [2 marks]

The Laplace transform of $t^2 e^{-2t}$ is:

(Numerical Answer Type — Enter the numerical value)

Q5. [2-MCQ] [2 marks]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

- (A) 0.2
- (B) 0.5
- (C) 0.6
- (D) 0.4

Q6. [2-MSQ] [2 marks]

Number of spanning trees in complete graph K_4 :

- (A) 4
- (B) 8
- (C) 16
- (D) 12

Q7. [2-MCQ] [2 marks]

Consider $f(x) = |x|$ on $[-1,1]$. This function is:

- (A) Differentiable everywhere
- (B) Continuous but not differentiable at 0
- (C) Neither continuous nor differentiable
- (D) Differentiable at 0 only

Q8. [2-NAT] [2 marks]

The differential equation $y \, dx - x \, dy = 0$ has solution:

(Numerical Answer Type — Enter the numerical value)

Q9. [2-NAT] [2 marks]

Gauss-Seidel method converges when the coefficient matrix is:

(Numerical Answer Type — Enter the numerical value)

Q10. [2-MCQ] [2 marks]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

- (A) 0.908
- (B) 0.092
- (C) 0.762
- (D) 0.195

Q11. [2-MCQ] [2 marks]

If $f(x) = x^3 - 3x + 2$, number of real roots:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q12. [2-MCQ] [2 marks]

Which of the following is a probability generating function for $\text{Poisson}(\lambda)$?

- (A) $e^{\lambda(z-1)}$
- (B) $1/(1-z)$
- (C) $\ln(1-z)$
- (D) $e^{(-\lambda)(1+z)}$

Q13. [2-MCQ] [2 marks]

For the ODE $dy/dx = y/x + \sqrt{y^2/x^2}$, with $y(1)=0$, the solution is:

- (A) $y = 0$
- (B) $y = x \ln(x)$
- (C) $y = x \sin(\ln x)$
- (D) $y = x^2$

Q14. [2-MCQ] [2 marks]

Rolle's theorem requires $f(x)$ to be:

- (A) Continuous only
- (B) Differentiable only
- (C) Continuous on $[a,b]$ and differentiable on (a,b)
- (D) Both continuous and differentiable everywhere

Q15. [2-MCQ] [2 marks]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq$:

- (A) $\|u\| * \|v\|$
- (B) $\|u\| + \|v\|$
- (C) $\|u\| / \|v\|$
- (D) $\sqrt{\|u\| * \|v\|}$

Q16. [2-MSQ] [2 marks]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\text{trace}(A^2)$ equals:

- (A) 14
- (B) 6
- (C) 36
- (D) 12

Q17. [2-MCQ] [2 marks]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

- (A) 1
- (B) 0
- (C) -1
- (D) $\cos(\pi)$

Q18. [2-MCQ] [2 marks]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

- (A) e
- (B) e^2
- (C) 1
- (D) infinity

Q19. [2-NAT] [2 marks]

Laplace transform of $e^{-3t}u(t)$ is:

(Numerical Answer Type — Enter the numerical value)

Q20. [2-MCQ] [2 marks]

If $f(z) = z^2 + 3z + 2$ is analytic, then $\text{Re}(f(i)) =$

- (A) -1
- (B) -2
- (C) 1

(D) 2

Q21. [2-MCQ] [2 marks]

For matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $A^{50} * [1, 0]^T$ equals:

- (A) $[1, 0]^T$
- (B) $[2, 1]^T$
- (C) $[(3+2\sqrt{2})^{50}, \dots]$
- (D) Cannot determine

Q22. [2-MCQ] [2 marks]

The rank of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$ is:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q23. [2-NAT] [2 marks]

Fourier series of $f(x)=x$ on $(-\pi, \pi)$ has only:

(Numerical Answer Type — Enter the numerical value)

Q24. [2-NAT] [2 marks]

For a Poisson distribution with mean 4, variance is:

(Numerical Answer Type — Enter the numerical value)

Q25. [2-NAT] [2 marks]

$P(|Z| < 1)$ for $Z \sim N(0, 1)$ is approximately:

(Numerical Answer Type — Enter the numerical value)

Q26. [2-NAT] [2 marks]

The area of a circle of radius r is:

(Numerical Answer Type — Enter the numerical value)

Q27. [2-MCQ] [2 marks]

The vector $(1, 2, 3)$ projected onto $(2, 1, 0)$ is:

- (A) $(\frac{4}{5}, \frac{2}{5}, 0)$
- (B) $(\frac{8}{5}, \frac{4}{5}, 0)$
- (C) $(2, 1, 0)$
- (D) $(1, 2, 3)$

Q28. [1-NAT] [1 mark]

Strongest acid:

(Numerical Answer Type — Enter the numerical value)

Q29. [2-MCQ] [2 marks]

pH of 10^{-8} M HCl approx:

- (A) 6
- (B) 7
- (C) 8
- (D) 6.98

Q30. [2-MCQ] [2 marks]

$K_p = K_c(RT)^{\Delta n}$. For $N_2 + 3H_2 \rightleftharpoons 2NH_3$, $\Delta n =$

- (A) 2
- (B) -2
- (C) 1
- (D) -1

Q31. [2-MCQ] [2 marks]

First-order half-life with $k = 0.693 \text{ min}^{-1}$:

- (A) 1 min
- (B) 0.693 min
- (C) 1.44 min
- (D) 0.5 min

Q32. [2-MCQ] [2 marks]

Galvanic cell converts:

- (A) Electrical to chemical
- (B) Chemical to electrical
- (C) Heat to work
- (D) Mechanical to electrical

Q33. [1-NAT] [1 mark]

Fixed-free Euler buckling:

(Numerical Answer Type — Enter the numerical value)

Q34. [1-MCQ] [1 mark]

Cantilever UDL w over L . Deflection:

- (A) $wL^4/(8EI)$
- (B) $wL^4/(3EI)$
- (C) $5wL^4/(384EI)$
- (D) $PL^3/(3EI)$

Q35. [2-NAT] [2 marks]

Max shear stress in solid shaft:

(Numerical Answer Type — Enter the numerical value)

Q36. [2-MCQ] [2 marks]

SS beam UDL w N/m, span L . Max BM:

- (A) $wL^2/8$
- (B) $wL^2/12$
- (C) $wL^2/4$
- (D) $wL^2/2$

Q37. [2-NAT] [2 marks]

Moment distribution DF for $K_{AB} = 4EI/L$, $K_{AC} = 3EI/L$:

(Numerical Answer Type — Enter the numerical value)

Q38. [2-MCQ] [2 marks]

Influence line for SS beam reaction:

- (A) Triangle

- (B) Rectangle
- (C) Linear decreasing
- (D) Parabola

Q39. [2-NAT] [2 marks]

Cantilever UDL w, L. Fixed end moment:

(Numerical Answer Type — Enter the numerical value)

Q40. [1-NAT] [1 mark]

Manning's equation for:

(Numerical Answer Type — Enter the numerical value)

Q41. [1-NAT] [1 mark]

Gumbel distribution used for:

(Numerical Answer Type — Enter the numerical value)

Q42. [1-MCQ] [1 mark]

Rainfall measured by:

- (A) Hygrometer
- (B) Rain gauge
- (C) Anemometer
- (D) Barometer

Q43. [1-NAT] [1 mark]

Darcy-Weisbach for:

(Numerical Answer Type — Enter the numerical value)

Q44. [1-MCQ] [1 mark]

Sum of first n odd numbers:

- (A) n^2
- (B) $2n$
- (C) $n(n+1)$
- (D) $n(n-1)$

Q45. [1-MCQ] [1 mark]

Compound interest: P at r% for 2 years =

- (A) $P(1+r/100)^2$
- (B) $P(1+2r/100)$
- (C) $P + Pr/100 \times 2$
- (D) $P \times r/100$

Q46. [2-MCQ] [2 marks]

If $x + 1/x = 3$, then $x^3 + 1/x^3 =$

- (A) 18
- (B) 27
- (C) 9
- (D) 3

Q47. [1-MCQ] [1 mark]

Slump test measures:

- (A) Strength
- (B) Workability
- (C) Durability
- (D) Setting time

Q48. [1-MCQ] [1 mark]

Camber on road provides:

- (A) Speed
- (B) Drainage
- (C) Aesthetics
- (D) Strength

Q49. [2-NAT] [2 marks]

M25 concrete modulus of elasticity:

(Numerical Answer Type — Enter the numerical value)

Q50. [1-MCQ] [1 mark]

Moment-area method gives:

- (A) Slope
- (B) Deflection
- (C) Both
- (D) None

Q51. [1-MCQ] [1 mark]

Method of joints gives:

- (A) Forces only
- (B) Moments
- (C) Both
- (D) None

Q52. [1-MCQ] [1 mark]

Bending moment in UDL simply supported:

- (A) $wL^2/8$
- (B) $wL^2/12$
- (C) $wL/2$
- (D) wL^2

Q53. [1-NAT] [1 mark]

Rankine K_a for $\phi=30^\circ$:

(Numerical Answer Type — Enter the numerical value)

Q54. [1-NAT] [1 mark]

Terzaghi q_{ult} for $c=0$, $\phi=30^\circ$, $B=2\text{m}$, $D_f=1\text{m}$, $\gamma=18$:

(Numerical Answer Type — Enter the numerical value)

Q55. [1-MCQ] [1 mark]

LL=60%, PL=30%. PI =

- (A) 30

- (B) 60
- (C) 90
- (D) 30%

Q56. [1-MCQ] [1 mark]

Bearing capacity of footing:

- (A) Depends on soil
- (B) Depends on load
- (C) Both
- (D) None

Q57. [1-MCQ] [1 mark]

Atterberg limits:

- (A) Liquid limit
- (B) Plastic limit
- (C) Shrinkage
- (D) All

Q58. [1-NAT] [1 mark]

BOD stands for:

(Numerical Answer Type — Enter the numerical value)

Q59. [1-MCQ] [1 mark]

Sedimentation removes:

- (A) Dissolved
- (B) Suspended
- (C) Colloids
- (D) Gases

Q60. [1-MCQ] [1 mark]

PCU stands for:

- (A) A
- (B) B
- (C) C
- (D) D

Q61. [1-MCQ] [1 mark]

Camber provided for:

- (A) Strength
- (B) Drainage
- (C) Smoothness
- (D) Aesthetics

Q62. [1-MCQ] [1 mark]

Theodolite measures:

- (A) Distance
- (B) Angle
- (C) Level
- (D) All

Q63. [1-MCQ] [1 mark]

Dumpy level used for:

- (A) Angle
- (B) Distance
- (C) Levelling
- (D) All

Q64. [1-MCQ] [1 mark]

Water-cement ratio affects:

- (A) Strength only
- (B) Workability only
- (C) Both
- (D) None

Q65. [1-MCQ] [1 mark]

Initial setting time OPC:

- (A) 30 min
 - (B) 45 min
 - (C) 60 min
 - (D) 24 hr
-

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1MCQ	A	1
2	1MCQ	B	1
3	1MCQ	B	1
4	1MCQ	C	1
5	1MCQ	A	1
6	1MCQ	A	1
7	1MCQ	C	1
8	1MCQ	D	1
9	1NAT	B	1
10	1NAT	B	1
11	1MCQ	B	1
12	1NAT	B	1
13	1NAT	A	1

14	1MCQ	B	1
15	1MCQ	A	1
16	1MCQ	B	1
17	1MCQ	B	1
18	1MCQ	C	1
19	1MCQ	C	1
20	1MCQ	A	1
21	1NAT	A	1
22	1MCQ	A	1
23	1NAT	A	1
24	1MCQ	A	1
25	1NAT	B	1
26	1MCQ	B	1
27	1NAT	D	1
28	1MCQ	B	1

29	1MCQ	A	1
30	1MCQ	B	1
31	2NAT	A	2
32	2NAT	A	2
33	2MCQ	B	2
34	2NAT	A	2
35	2NAT	A	2
36	2MCQ	D	2
37	2MCQ	A	2
38	2MSQ	C	2
39	2MCQ	B	2
40	2MCQ	C	2
41	2MCQ	A	2
42	2MCQ	B	2
43	2NAT	B	2

44	2NAT	C	2
45	2MCQ	A	2
46	2MCQ	B	2
47	2MCQ	A	2
48	2MCQ	A	2
49	2MCQ	C	2
50	2NAT	A	2
51	2MCQ	A	2
52	2MCQ	A	2
53	2MSQ	A	2
54	2MCQ	B	2
55	2MCQ	C	2
56	2MCQ	B	2
57	2NAT	A	2
58	2MCQ	A	2

59	2MCQ	A	2
60	2MCQ	A	2
61	2NAT	A	2
62	2NAT	A	2
63	2NAT	A	2
64	2NAT	A	2
65	2MCQ	B	2

DETAILED SOLUTIONS

Q1.

Answer: A

Solution:

Correct Answer: A

Subject: General Aptitude | Topic: Quantitative | Difficulty: moderate

Q2.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Verbal | Difficulty: easy

Q3.

Answer: B

Solution:

Correct Answer: B

Subject: Engineering Mathematics | Topic: Differential Equations | Difficulty: moderate

Q4.

Answer: C

Solution:

Correct Answer: C

Subject: Structural Engineering | Topic: Deflection | Difficulty: moderate

Q5.

Answer: A

Solution:

Correct Answer: A

Subject: Structural Engineering | Topic: Trusses | Difficulty: moderate

Q6.

Answer: A

Solution:

Correct Answer: A

Subject: Structural Engineering | Topic: Beams | Difficulty: difficult

Q7.

Answer: C

Solution:

Correct Answer: C

Subject: Geotechnical Engineering | Topic: Foundation | Difficulty: difficult

Q8.

Answer: D

Solution:

Correct Answer: D

Subject: Geotechnical Engineering | Topic: Soil Mechanics | Difficulty: moderate

Q9.

Answer: B

Solution:

Correct Answer: B

Subject: Water Resources Engineering | Topic: Open Channel | Difficulty: easy

Q10.

Answer: B

Solution:

Correct Answer: B

Subject: Water Resources Engineering | Topic: Flood | Difficulty: moderate

Q11.

Answer: B

Solution:

Correct Answer: B

Subject: Water Resources Engineering | Topic: Precipitation | Difficulty: easy

Q12.

Answer: B

Solution:

Correct Answer: B

Subject: Water Resources Engineering | Topic: Pipe Flow | Difficulty: moderate

Q13.

Answer: A

Solution:

Correct Answer: A

Subject: Environmental Engineering | Topic: Sewage | Difficulty: easy

Q14.

Answer: B

Solution:

Correct Answer: B

Subject: Environmental Engineering | Topic: Water | Difficulty: difficult

Q15.

Answer: A

Solution:

Correct Answer: A

Subject: Transportation Engineering | Topic: Traffic | Difficulty: difficult

Q16.

Answer: B

Solution:

Correct Answer: B

Subject: Transportation Engineering | Topic: Highways | Difficulty: moderate

Q17.

Answer: B

Solution:

Correct Answer: B

Subject: Surveying & Geomatics | Topic: Theodolite | Difficulty: difficult

Q18.

Answer: C

Solution:

Correct Answer: C

Subject: Surveying & Geomatics | Topic: Levelling | Difficulty: easy

Q19.

Answer: C

Solution:

Correct Answer: C

Subject: Construction Materials | Topic: Mix Design | Difficulty: moderate

Q20.

Answer: A

Solution:

Correct Answer: A

Subject: Construction Materials | Topic: Cement | Difficulty: difficult

Q21.

Answer: A

Solution:

$K_a = \tan^2(45-15) = \tan^2(30) = 1/3.$

Subject: Geotechnical | Topic: Geotechnical | Difficulty: difficult

Q22.

Answer: A

Solution:

$$1+3+5+\dots+(2n-1) = n^2.$$

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: moderate

Q23.

Answer: A

Solution:

$$q_{ult} = q \cdot N_q + 0.5 \cdot \gamma \cdot B \cdot N_\gamma = 18 \cdot 22.5 + 0.5 \cdot 18 \cdot 2 \cdot 19.7 = 405 + 354.6 = 759.6 \text{ kPa}.$$

Subject: Geotechnical | Topic: Geotechnical | Difficulty: difficult

Q24.

Answer: A

Solution:

$$PI = LL - PL = 60 - 30 = 30.$$

Subject: Geotechnical | Topic: Geotechnical | Difficulty: moderate

Q25.

Answer: B

Solution:

$$K=2, L_{eff}=2L. P_{cr} = \pi^2 EI / (2L)^2 = \pi^2 EI / (4L^2).$$

Subject: SOM | Topic: SOM | Difficulty: moderate

Q26.

Answer: B

Solution:

Slump: workability/consistency of fresh concrete.

Subject: Civil Engineering | Topic: Civil Engineering | Difficulty: easy

Q27.

Answer: D

Solution:

Perchloric acid (HClO₄) is the strongest among common acids.

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q28.

Answer: B

Solution:

Camber: cross slope for rainwater drainage.

Subject: Civil Engineering | Topic: Civil Engineering | Difficulty: moderate

Q29.

Answer: A

Solution:

$$CI = P(1+r/100)^n. \text{ For 2 years: } P(1+r/100)^2.$$

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: moderate

Q30.

Answer: B

Solution:

$$\text{Cantilever UDL: } \delta = wL^4 / (8EI).$$

Subject: SOM | Topic: SOM | Difficulty: easy

Q31.

Answer: A

Solution:

$$\tau_{max} = T \cdot c / J = 16T / (\pi \cdot d^3).$$

Subject: SOM | Topic: SOM | Difficulty: moderate

Q32.

Answer: A

Solution:

$$L\{f(t)\} = -d/ds(F(s)). F(s) = 1/(s+2). -d/ds[1/(s+2)] = 1/(s+2)^2.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q33.

Answer: B

Solution:

$$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q34.

Answer: A

Solution:

$$E_c = 5000 \cdot \sqrt{f_{ck}} = 5000 \cdot 5 = 25000 \text{ MPa}.$$

Subject: Civil Engineering | Topic: Civil Engineering | Difficulty: moderate

Q35.

Answer: A

Solution:

$$DF_{AB} = 4/(4+3) = 4/7 = 0.57.$$

Subject: Structural Analysis | Topic: Structural Analysis | Difficulty: easy

Q36.

Answer: D

Solution:

Very dilute: $[H^+]$ from water dominates. $[H^+]_{\text{total}}$ approx 1.1×10^{-7} . pH approx 6.96.

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q37.

Answer: A

Solution:

$$\text{SS UDL: max BM at center} = wL^2/8.$$

Subject: SOM | Topic: SOM | Difficulty: moderate

Q38.

Answer: C

Solution:

$$\text{Kirchhoff: for } K_n, \text{ spanning trees} = n^{n-2} = 4^2 = 16.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q39.

Answer: B

Solution:

$$\Delta_n = 2 - 4 = -2.$$

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q40.

Answer: C

Solution:

$$R_A = (L-x)/L. \text{ Linear decreasing from } L \text{ to } 0.$$

Subject: Structural Analysis | Topic: Structural Analysis | Difficulty: easy

Q41.

Answer: A

Solution:

$$\text{Cube: } (x+1/x)^3 = x^3 + 3x + 3/x + 1/x^3 = 27. \quad x^3 + 1/x^3 = 27 - 3 \cdot 3 = 18.$$

Subject: Miscellaneous | Topic: Miscellaneous | Difficulty: easy

Q42.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q43.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C \cdot x$. Wait: $y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C \cdot x$. That's option A. Hmm, let me check: $y \, dx - x \, dy = 0 \Rightarrow y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow y = C \cdot x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q44.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q45.

Answer: A

Solution:

$P(X \geq 2) = 1 - e^{-4}(1+4) = 1 - 5e^{-4} \approx 0.908$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q46.

Answer: B

Solution:

$f(0)=2, f(1)=0, f(-1)=4, f(2)=4, f(-2)=-8+6+2=0$. Roots at $x=1$ (double?) and $x=-2$. $f(x)=(x-1)^2(x+2)$. Two distinct real roots.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q47.

Answer: A

Solution:

PGF of Poisson(λ): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q48.

Answer: A

Solution:

Homogeneous ODE. Substituting $y=x*v$ and $y(1)=0$ gives $y=0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q49.

Answer: C

Solution:

Rolle's: f continuous on $[a,b]$, differentiable on (a,b) , $f(a)=f(b)$. Then $f'(c)=0$ for some c in (a,b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q50.

Answer: A

Solution:

$FEM = wL*(L/2) = wL^2/2$ (hogging).

Subject: Structural Analysis | Topic: Structural Analysis | Difficulty: easy

Q51.

Answer: A

Solution:

$t_{1/2} = 0.693/k = 0.693/0.693 = 1$ min.

Subject: Chemistry | Topic: Chemistry | Difficulty: easy

Q52.

Answer: A

Solution:

$|\langle u, v \rangle| \leq \|u\| * \|v\|$. Equality when u and v are linearly dependent.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q53.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1+4+9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q54.

Answer: B

Solution:

Galvanic: spontaneous redox converts chemical to electrical energy.

Subject: Chemistry | Topic: Chemistry | Difficulty: moderate

Q55.

Answer: C

Solution:

$$y = A \cos(2x) + B \sin(2x). A=1, B=0. y(\pi/2) = \cos(\pi) = -1.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q56.

Answer: B

Solution:

$$(1+1/n)^{2n} = [(1+1/n)^n]^2 \rightarrow e^2 \text{ as } n \rightarrow \infty.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q57.

Answer: A

Solution:

$$L\{e^{-at}\} = 1/(s+a). \text{ Here } a=3: 1/(s+3).$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q58.

Answer: A

Solution:

$$f(i) = i^2 + 3i + 2 = -1 + 3i. \operatorname{Re}(f(i)) = -1.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q59.

Answer: A

Solution:

$$\text{Eigenvalues are 3 and 1. } [1,0]^T \text{ is eigenvector for eigenvalue 1. } A^{50} [1,0]^T = [1,0]^T.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q60.

Answer: A

Solution:

$$\text{Row 2} = 2 \times \text{Row 1}, \text{Row 3} = 3 \times \text{Row 1}. \text{ All rows linearly dependent. Rank} = 1.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q61.

Answer: A

Solution:

$$f(x)=x \text{ is odd on symmetric interval. Fourier series has only sine terms.}$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q62.

Answer: A

Solution:

$$\text{Poisson: mean} = \lambda = \text{variance. Both} = 4.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q63.

Answer: A

Solution:

$$\text{Standard normal: } P(-1 < Z < 1) \approx 0.68.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q64.

Answer: A

Solution:

$$\text{Area} = \pi r^2. \text{ Circumference} = 2\pi r.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q65.

Answer: B

Solution:

$$\operatorname{proj} = ((v \cdot w) / (w \cdot w)) * w = (4/5) * (2, 1, 0) = (8/5, 4/5, 0).$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

PAPER BLUEPRINT

Subject Distribution:

Engineering Mathematics: 25 questions, 49 marks
Chemistry: 5 questions, 9 marks
SOM: 4 questions, 6 marks
Structural Analysis: 3 questions, 6 marks
Water Resources Engineering: 4 questions, 4 marks
Miscellaneous: 3 questions, 4 marks
Civil Engineering: 3 questions, 4 marks
Structural Engineering: 3 questions, 3 marks
Geotechnical: 3 questions, 3 marks
General Aptitude: 2 questions, 2 marks
Geotechnical Engineering: 2 questions, 2 marks
Environmental Engineering: 2 questions, 2 marks
Transportation Engineering: 2 questions, 2 marks
Surveying & Geomatics: 2 questions, 2 marks
Construction Materials: 2 questions, 2 marks

Difficulty Distribution:

Easy: 19 | Moderate: 36 | Difficult: 10

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.